Agentic Toolkits for Architects & Process Engineers

Agents to assist with analysis, design & delivery

What are the main activities that a senior consultant does…

A consultant often gets involved in many different architecture, design & process streams. As such, the main type of activities that they do are…

Requirements Management – Review an organisation's needs & create requirements that capture those needs

Process Engineering – Review an organisation's value streams or business flows & design processes that can formalise or improve those flows

Architecture – Review an organisation's system's requirements & design (or refine) eco-systems that meet those needs

System Design – Taking architectural overviews & create solid high-level & low-level design assets to deliver them

Critical Reviews – Review existing process, architectural & design assets to identify gaps, improvements or changes needed to make them fit for purpose now & in the future



Wouldn't it be cool if AI could help with these activities in a more constructive way? Well…

Agentic Toolkits Created or Optimised for the Hackathon…

Three main streams of agentic toolkits have been created to support senior consulting workstreams.

Cloud Architecture

- Import requirements from PDFs, emails, UI attachments & others
- Create architectural diagrams
- Simulate “red teams” to attack that architecture
- Use FinOps to cost & optimize resource use
- Run operational simulations to identify issues
- Ask "what-if” questions
- Update that architecture with specific requests

Architecture & Design

- Import requirements (see Cloud architecture)
- Importing existing HLDs/LLDs
- Create HLDs, LLDs & supporting design assets
- Run operational simulations to identify issues
- Ask "what-if” questions
- Update designs with specific requests
- Ground designs in contextual sources like Wiki’s & RAGs where REST APIs are supported

Process Engineering

- Import requirements (see previous)
- Importing existing process assets
- Create process documentation & supporting assets
- Run operational simulations to identify issues
- Ask "what-if” questions
- Update processes with specific requests
- Ground processes in contextual sources like Wiki’s & RAGs where REST APIs are supported

Outstanding issues…

The following are outstanding issues or improvements that are needed

Improvement	Description
Documentation formatting	• Some of the documents created need formatting improvements as document flow could read better
Intent analysis	• Sometimes agents get confused by vague intent and get directed to “general consulting” agents, rather than more specialised agents
Diagram Generation	• Diagram formatting for Cloud architecture could do with some improvements
HLD/LLD Split documents	• Individual document generation for HLD and LLD documents should be done
Production Improvements	• Outstanding code refactoring for code reuse and optimisations • Add context guardrails to protect against prompt injection and context manipulation